

Tópicos de Economía de Género

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Maestría en Economía

Clase 5 (Parte 1)

Segregación y brechas salariales

- Segregación ocupacional y brechas de género en ingresos laborales
 - **Factores de oferta:** diferencias en capital humano, experiencia, habilidades, preferencias, motivación, expectativas, entre otros factores que analizaremos hoy
 - **Factores de demanda:** discriminación

Factores de demanda: Discriminación

Discrimination

(Unacceptable) discrimination in the labor market:

- if workers with the same characteristics in the same labor market are treated differently in terms of:
 - hiring
 - wage
 - promotion decisions (glass ceiling)
- i.e. unequal **treatment** of workers who **differs in terms of their non-productive characteristics** (e.g. gender, race/nationality, religion, physical appearance, sexual preferences, etc).

*"Over the last couple of decades, a rich literature has leveraged experiments (in the lab and in the field) to provide **convincing evidence that discrimination, thus defined, indeed exists**" (Bertrand and Duflo, 2017)*

Not discrimination

Two workers living in the same city at the same time but having different wages is not necessarily evidence of unacceptable discrimination.

Some determinants of wage differentials are widely accepted:

- Variation in characteristics related to **productivity**
 - **acquired** human capital: returns to schooling, experience, skills, etc.
 - **innate**: intelligence, non-cognitive abilities, etc.
- Variation in **supply/demand** for specific jobs
 - e.g. **compensating wage** differential for jobs that have unattractive attributes as health risks

Theory

- Two workhorse models of discrimination in the economics literature
 - **taste-based discrimination**
 - **statistical discrimination**

Taste-based explanation for discrimination

Becker (1957): some employers have a **distaste** (disutility) for hiring members of the minority group.

- They may indulge this distaste by refusing to hire them, or if they do hire them, paying them less than other employees for the same level of productivity
- If the **fraction of discriminating employers** in the economy is sufficiently large, a wage differential will emerge in equilibrium between otherwise identically productive minority and majority employees
- By electing to not hire minority workers, **prejudiced employers will experience lower profits**,
 - under perfect **competition**, discriminating employers would be wiped away and **taste-based discrimination would disappear**

Taste-based discrimination

Denote “a” as majority group and “b” as minority group membership

Employers maximize

$$U = pF(N_a + N_b) - w_a N_a - w_b N_b - d N_b$$

- F is the production function
- p is the price of goods
- N_j is the # of workers of group $j \in \{a; b\}$
- w_j is the wage payed to workers of group $j \in \{a; b\}$
- $d \geq 0$ is the distaste parameter of the employer towards member of group b

Prejudiced employers ($d > 0$) will act as the wage of workers belonging to group b was $w_b + d$

- they will only hire workers of group b if: $w_a \geq w_b + d$

Taste-based discrimination: Example

- Each firm wants to hire one worker (# of firms, N_f , equals # of workers)
 - $N_f = N_a + N_b$
- Each firm is willing to pay $w_a = w$ to workers of group a
- A share α of firms is not prejudiced and thus willing to pay $w_b = w$
- The share $(1 - \alpha)$ of firms has a prejudice parameter of $d > 0$ and is only willing to hire workers of group b at the wage $w_b = w_a - d < w$
- Then, **workers of group b will seek employers who are not prejudiced**

Taste-based discrimination: Example

Equilibrium Wages

- If $\alpha N_f \geq N_b$ (enough unprejudiced employers to hire all workers of group b)
 - In equilibrium, $w_a = w_b = w$
- If $\alpha N_f < N_b$ (some workers of group b will end up at prejudiced employers)
 - wages are determined by the marginal firm: $w_a = w$ and $w_b = w - d$

Wage depend on the relative share of minorities among workers, and of unprejudiced employers among firms

- **the existence of prejudiced employers does not cause discrimination in equilibrium as long as the marginal employer is unprejudiced**

Taste-based discrimination: Competition

Competition should eradicate discrimination:

- In an equilibrium with taste-based wage discrimination ($\alpha N_f < N_b$), **non-discriminating employers make positive profits**
- With free market entry/exit, these positive profits **will should attract market entry/expansion of non discriminating firms**
 - Once the share of non discriminating firms becomes sufficiently large, discrimination will be eliminated

Why taste-based discrimination may persist

Taste-based employer discrimination can persist if:

- ① labor markets are **monopsonistic**
- ② workers face **search costs & imperfect information** of employers' tastes
 - In equilibrium a firm will offer a worker their productivity minus the worker's search cost (Black, JoLE 1995)
 - Because minority workers face a higher risk of not getting an offer back, firms will offer them lower wages at equilibrium
 - As long as there are a few prejudiced employers and imperfect information, minority workers will receive a lower salary in equilibrium
- ③ **customer discrimination:**
 - customers might have preferences for the type of worker with whom they interact in shops, restaurants (e.g. waiters' race/gender/physical appearance)
 - employers who observe their customers care about certain characteristics of their employees will have the incentive to hire according to those preferences
 - differences in productivity?: discriminating customers make race/ gender/ appearance a productive characteristic as it contributes to the profit of the firm and sales performance!

Statistical Discrimination

Differential treatment of members of the minority group **is due to imperfect information**, and discrimination is the result of a signal extraction problem.

(Phelps, 1972; Arrow, 1973; Aigner and Cain, 1977)

Statistical discrimination

It's plausible that **employers cannot perfectly assess worker productivity** (particularly at the time of hire) and instead make educated guesses.

- firms will **use easily observable characteristics such as race or gender to infer the expected productivity** of applicants if these characteristics are correlated with productivity.

E.g. it may be known to employers that minority applicants are on average less productive than majority applicants,

- then, an employer who sees two applicants with similar noisy but unbiased signals of productivity should rationally favor the majority applicant to the minority one as her **expected productivity** is higher.
- While **expected productivity will equal true productivity on average within each group**, statistical discrimination will result in some minority workers being treated less favorably than non-minority workers of the same true productivity.

Statistical discrimination

- Two groups of applicants (female/male, White/Black, ...), $j \in \{a; b\}$
- Applicant have different levels of the value of the marginal product, VMP_i
- **Employers cannot directly observe the productivity of job applicants.** Therefore, they give workers a standardized test (or **check their grades** from school, or employ them for a trial period)
 - denote T_i the result of the test for individual i
- The **test is a noisy** (but unbiased) **predictor** of VMP_i
- **Employers observe some statistical information** (hence, statistical discrimination): $\overline{VMP_a}$ and $\overline{VMP_b}$
 - If group belonging is correlated with average productivity, it provides an additional, imperfect signal for applicants' productivity.

Interpreting the signal

- Employers will now estimate the expected marginal productivity of the candidate, given the signal provided by the test:

- $E(VMP_{ij} \mid T_i; \overline{T_j}) = \alpha T_i + (1 - \alpha) \overline{VMP_j}$

- When the test provide a precise signal, $\alpha \rightarrow 1$
- When the test is uninformative, employers rely on their priors, $\alpha = 0$

Even with competition:

- Workers of each group **are on average paid according to their true expected productivity**
- **For equally productive workers, workers of group b are paid less** than workers of group a
($\overline{VMP_a} > \overline{VMP_b}$)

Efficiency of statistical discrimination

While taste-based discrimination is clearly inefficient (simply consider how it constrains the allocation of talent), **statistical discrimination is efficient**

- Therefore, many economists would say that employers 'should' statistically discriminate because it is **profit-maximizing, not motivated by animus**, and arguably even '**fair**' since it treats people with the same expected productivity equally

Despite economic 'efficiency', statistical discrimination is often **unlawful**

- Statistical discrimination is difficult to detect, however, and so it is plausible that it occurs frequently despite the law.

Self-fulfilling prophecies

Whether discrimination is taste-based or statistical, it may ultimately result in genuine difference between groups, through **self-fulfilling prophecies**

- If the stereotypical woman is not good at math, talented girls may **become discouraged** and ultimately not become good at math
- If teachers or employers assume that students of a particular color are less smart, they will invest less in them

Discrimination that starts as taste-based and inefficient can easily morph into the more “justifiable” form

- Due to expected discrimination, the difference in actual average human capital increases
- “Valid” stereotypes today could be the product of ambient animus

Measuring Discrimination

- Tempting to take these descriptive statistics as evidence of discrimination
- However, gender gap in average earnings can be due to a combination of factors as different human capital levels, job characteristics, working profile (even it could be due to discouragement due to pessimistic expectations)

Earlier research on discrimination focused on individual-level outcome regressions using observational data

- To assess the the scope of potential discrimination, one could in principle estimate
$$Y = X\beta + \alpha Z + \varepsilon$$
 - where X is a vector of exogenous productivity characteristics and Z is an indicator variable for membership in a minority group
 - Assuming that X fully captures the set of productive characteristics and their returns, then discrimination occurs when $\alpha < 0$.
- discrimination was estimated from the **“minority” differential that remained unexplained after including as many proxies as possible for productivity**

Measuring Discrimination

Problems with using this regression approach

- **omitted variables bias:** we don't really know whether the workers with comparable observable characteristics indeed have comparable unobserved characteristics ($\text{Cov}(Z; \varepsilon) \neq 0$)
 - i.e. race or gender could be correlated with other proxies for productivity that are unobservable to the researcher but observed by the employer
- **is X truly exogenous?:** the variables the researcher controls for might themselves be affected by discrimination:
 - the X's could be endogenous due to **expectations** of discrimination (investment in schooling; **sorted** into industries where there is no or limited discrimination, etc)
 - disadvantaged groups may not have **access** to high-quality schools because of discrimination...
 - then, adding more and more controls to a regression could ultimately obscure the interpretation of the evidence
- **Solution:** e.g. **audit and correspondence** methodologies

Audit and Correspondence Studies

Field Experiments on Discrimination

(from Handbook of Field Experiments 2017, Chapter 8, by Bertrand and Duflo)

Experimental methods employed to **measure the prevalence of discrimination**

- **Audit studies**

- Audit studies send out individuals who are matched in all observable characteristics except for the one in question (race, criminal record, etc.) to apply for jobs or make purchases, then researchers analyze the responses

- **Correspondence studies**

- they represent by far the largest share of field experiments on discrimination – do the same but control for more variables by creating fictitious applicants for jobs who correspond via mail.

Audit Studies

- Two individuals are matched for all relevant personal characteristics other than the one that is presumed to lead to discrimination, e.g. race, ethnicity, gender.
- They then apply for a job
- The results they achieve and the treatment they receive in the transaction are closely observed, documented, and analyzed to determine if the outcomes reveal patterns of differential treatment
- Discrimination is said to have been detected when “auditors in the protected class are systematically treated worse than their teammates”

Limitations of audit studies

- Employ actors → high cost per observation → **low sample size**
- Unlikely to erase the numerous **differences** that exist between the auditors
- They are **not double-blind**: auditors know the purpose of the study: This may generate conscious or subconscious motives among auditors to generate data consistent or inconsistent with their beliefs about race or gender issues

Correspondence studies

In response to a job advertisement, the researcher sends (many) pairs of near-identical CVs, one of which is assigned the perceived minority trait.

Rely on fictitious applicants, thus address some weaknesses of audit studies

- one can be sure to generate strict comparability across groups for all information that is seen by the employers → any observed differences are caused solely by the minority trait manipulation
- size: the low marginal cost allows to send out a large number of applications
 - more precise estimates
 - heterogeneity analysis: allows researchers to examine the nature of the differential treatment from many more angles

Correspondence studies

Discrimination is estimated by comparing the outcomes for the fictitious applicants with and without the perceived minority trait

How to manipulate the perceived minority trait?

- e.g. names of the applicants (e.g. female names, African-American names, Arabic Names, etc).

Outcomes mainly limited to measuring **call-backs** by employers

Correspondence studies

Correspondence studies have by been carried in many countries around the world, and have focused on a variety of perceived traits that can be randomized on a CV

- The studies have tended to be fairly close replications of BM for different populations or contexts
 - *"The literature has failed to push the correspondence methodology to design approaches to more formally test for various theories of why differential treatment is taking place"*
(Bertrand and Duflo, 2017)

Correspondence studies

Summary of results from the empirical literature using correspondence studies:

- **Overwhelming evidence of discrimination in the labor market against racial** and ethnic minorities.
- **Fewer studies on gender**, and discrimination against women at the call-back stage is much less apparent in general.
 - Some studies attempt to show whether the degree (and nature) of discrimination depends on the nature of the profession: e.g. on female-dominated professions call-back is in favor of women (Booth and Leigh, 2010).
 - According to Bertrand and Duflo (2017), **a topic of interest for future work** would be to apply the correspondence method to measure the **role of family constraints in gender discrimination**
 - e.g. measure the extent to which a bias exists against women with children, or against young women who may have children in the future (as in Becker 2019).

Correspondence studies: Gender Discrimination in Hiring

Paper: Discrimination in Hiring Based on Potential and Realized Fertility: Evidence from a Large-Scale Field Experiment, Sascha O. Becker Ana Fernandes Doris Weichselbaumer, IZA DP No. 12308, 2019

- Debido a las normas de género convencionales, las mujeres tienen más probabilidades de estar a cargo del cuidado de los niños que los hombres
- Desde la perspectiva de un empleador, las mujeres en edad fértil también están en "riesgo" de embarazo
- Ambos factores potencialmente afectarían las decisiones de contratación de las empresas
- En este artículo, los autores enviaron por correspondencia aprox. 9,000 solicitudes de empleo (en Alemania, Suiza y Austria)
- Los CVs variaban en las características personales del candidato, como el estado civil y la edad de los hijos

Teoría

Examinan dos rutas por las que la maternidad puede dar lugar a discriminación estadística por parte de las empresas

- ① "Riesgo de **embarazo**" (desde la perspectiva de los empleadores) que cae 100 % en las mujeres, y no en los hombres
 - ① Esto conlleva costos para los empleadores que necesitan encontrar al menos un reemplazo temporal para las empleadas que dan a luz
- ② Las normas de género tradicionales asignan responsabilidades de **cuidado infantil** principalmente a las mujeres.
 - ① Esto hace, por ejemplo, que sea más probable que sean las mujeres quienes se tomen un día libre para cuidar a un niño enfermo, y así las empresas pueden creer que contratar a mujeres sea más costoso que a hombres «similares»
 - ② Entre las mujeres, aquellas con hijos mayores tendrían menos probabilidades de tener que lidiar con problemas de cuidado infantil, ya que los niños mayores están en la escuela y / o tienen menos probabilidades de sufrir problemas de salud que requieren la atención de los padres

Teoría

- Para probar estos potenciales efectos del posible embarazo y el cuidado de los niños, analizan cómo varían las probabilidades de contratación de mujeres que se encuentran en diferentes fases de su formación familiar
- Experimento: al candidato ficticio de 30 se les da al azar diversos datos demográficos del hogar:
 - 1 solteros y sin hijos
 - 2 casados y sin hijos
 - 3 casados y dos hijos pequeños
 - 4 casados y dos hijos mayores
 - 5 sin información sobre la composición del hogar

Los candidatos aplican a trabajos de secretaría y contabilidad, respondiendo a anuncios de trabajo publicados en internet. Las posiciones son:

- 1 full-time
- 2 part-time

Correspondence studies: Gender Discrimination in Hiring

- Resultados: para trabajos de medio tiempo, las mujeres casadas con hijos mayores, quienes probablemente terminaron su ciclo de maternidad y dedican menos tiempo a cuidar niños que las mujeres con niños muy pequeños, tienen una ventaja significativa en comparación con otros grupos de mujeres
- Las candidatas casadas, pero sin hijos, que tienen una mayor probabilidad de quedar embarazadas, se encuentran en desventaja en comparación con las solteras, pero sin hijos para trabajos de medio tiempo

Correspondence studies: Gender Discrimination in Hiring

Probabilidad de que las llamen para entrevista: Muestra de candidatas a trabajos part-time

	(1)	(2)
Married No Kids	-0.057* (0.030)	-0.052* (0.030)
Married 2 Young Kids	0.019 (0.035)	0.021 (0.034)
Married 2 Old Kids	0.084** (0.040)	0.082** (0.039)
No Info on Family Status	0.007 (0.034)	-0.008 (0.034)
City Dummies	no	yes
Time Dummies (Year and Quarter)	no	yes
Application Characteristics	no	yes
Firm Characteristics	no	yes
Observations	1,332	1,327
R-squared	0.012	0.102
Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Correspondence studies

- Estos efectos no están presentes para los trabajos de tiempo completo,
- quizás porque al solicitarlos las mujeres están señalando que están organizadas (externalizado) en cuanto al cuidado de sus hijos

Prob. de que las llamen para entrevista: Muestra de candidatas a trabajos full-time

	(1)	(2)
Married No Kids	-0.012 (0.018)	-0.014 (0.018)
Married 2 Young Kids	-0.022 (0.020)	-0.030 (0.020)
Married 2 Old Kids	-0.009 (0.020)	-0.005 (0.020)
No Info on Family Status	-0.007 (0.019)	-0.003 (0.019)
City Dummies	no	yes
Time Dummies (Year and Quarter)	no	yes
Application Characteristics	no	yes
Firm Characteristics	no	yes
Constant	0.206*** (0.014)	0.280*** (0.060)

Experimentos: lo que hemos aprendido

Paper: Gender and the labor market: What have we learned from field and lab experiments?, Ghazala Azmat y Barbara Petrongolo, Labour Economics 2014

- *«Si bien no es trivial extrapolar una visión de consenso clara de la literatura experimental sobre discriminación de género, es justo resumir esta literatura concluyendo que hay evidencia de una discriminación significativa contra las mujeres en puestos de alto rango o dominados por hombres (y discriminación contra los hombres en puestos de trabajo femeninos)»*
- *«En comparación con el enfoque de regresión, el enfoque experimental tiende a encontrar pruebas mucho más limitadas de discriminación contra las mujeres en el mercado. Diferentes resultados de los dos enfoques pueden ser impulsados por brechas sistemáticas en inobservables a favor de los hombres, lo que contaminaría la brecha inexplicable en los salarios en los análisis de regresión»*

Experimentos: lo que aún nos queda por saber (Ghazala Azmat y Barbara, 2014)

«Aún quedan dos preguntas con respecto a la evidencia experimental sobre discriminación

- ***La naturaleza*** del grado de discriminación observada (*¿gusto o estadística?*)
 - *Para testear esto se debería manipular la la información disponible sobre los participantes, que sería irrelevante en el caso de discriminación basada en el gusto, pero altamente valorada en caso de discriminación estadística*
- ***Link*** entre el grado de discriminación revelado por los experimentos (especialmente los estudios de correspondencia) y los resultados de interés del mercado laboral
 - *Los estudios de correspondencia típicamente dan una idea de las diferencias de género en las tasas de llamado a entrevista, pero el impacto de la discriminación en las diferencias salariales y laborales depende de lo que suceda en la etapa de oferta de trabajo (durante y después de la entrevista)»*

Future research (Bertrand and Duflo, 2017)

*While field experiments in the last decade have been instrumental in documenting the prevalence of discrimination, field experiments in the future decade should aim to play as large of a role in **isolating effective methods to combat it.***

(Bertrand and Duflo, 2017)

Natural Experiments

Natural Experiments: Test of sex-biased hiring

Claudia Goldin and Cecilia Rouse, "Orchestrating Impartiality: The Impact of 'Blind' Auditions on Female Musicians," American Economic Review (2000)

Can blind auditions eliminate discrimination?

- US symphony orchestras long conducted non-blind auditions (musicians used to be handpicked by the music director)
- This hiring process typically led to a symphony orchestra that was composed of mostly male musicians (90 out of 100 musicians were male)
- The major orchestras adopted a process of "blind" auditions in the 1980s and 1990s.
- Applicants for a position at the orchestra would play a musical piece behind a screen (a piece of heavy cloth hanging from the ceiling).
- Persons involved in the hiring decision could hear the applicant play but could not see who the applicant was.

Natural Experiments: Test of sex-biased hiring

This paper takes advantage of this natural experiment

- Examine blind auditions in DiD framework
- Use data from actual auditions, in an individual fixed-effects framework

Results

- Blind audition procedure fostered impartiality in hiring and increased the proportion women in symphony orchestras
- It increased the probability that a female musician advanced out of the preliminary rounds by 50 percent.

Discriminación: Resumen de la literatura

En cuanto a la **contratación**, hay evidencia en parte consistente con discriminación basada en estereotipos de género en los trabajos:

- Discriminación contra las mujeres en los campos dominados por los hombres, y también en sectores mejor pagados
- Evidencia de discriminación contra las mujeres en trabajos dominados por hombres y contra hombres en trabajos dominados por mujeres

Discriminación hacia mujeres con responsabilidades familiares:

- Evidencia de discriminación contra las mujeres jóvenes basada en la futura maternidad esperada
- Evidencia de discriminación contra las mujeres con responsabilidades de crianza

Discriminación: Percepciones

Probabilidad de discriminación laboral por sexo (Latinobarómetro 2008)

